RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

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Mysuru Road Bengaluru - 560 059

III Semester B.E. Regular / Supplementary Examination February/March 2026

ELECTRONICS AND COMMUNICATION ENGINEERING

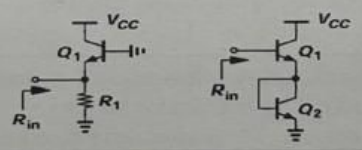
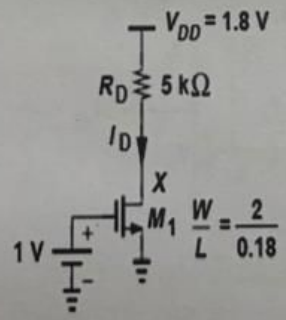
Analog Microelectronic Circuits

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

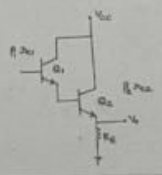
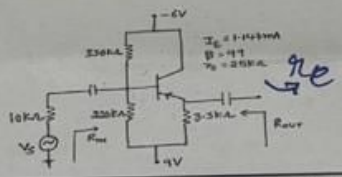
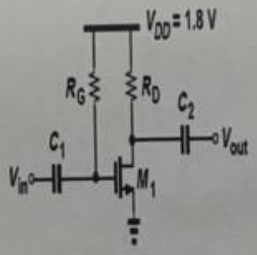
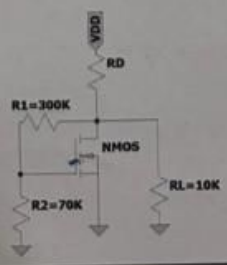
1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

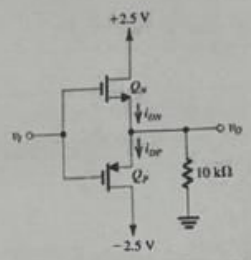
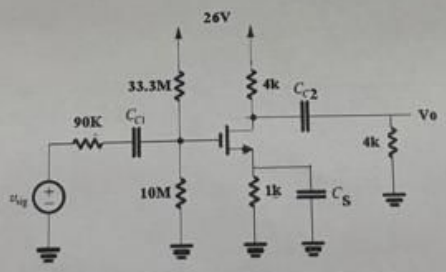
Question No	Question	M	CO	BT
1.1	Calculate the unity gain frequency F_T of a BJT with g_m = 1mA/V, C_{bc} = 0.1pF and C_{be} = 5pF.	02	1	1
1.2	Find the input resistances of the circuits shown in Fig. Assume I_E=1mA, beta=99. 	02	4	4
1.3	The change in drain voltage when the gate voltage increases by 10mV in the circuit shown in the figure below is ____mV. Assume transistor is in saturation with μnCox = 100μ A/V^2, Vt = 0.4 V, I_D=200μA 	02	4	5
1.4	The region of operation of the NMOS with V_G = 5V, V_S = 4V, V_D = 6V and V_t=2V is _____.	02	3	2
1.5	An NMOS has Kn=2 mA/V^2, Vt=1V and VGS=3V. Find gm and ro if channel-length modulation λ=0.02 V^{-1} and VDS=5V.	02	2	2
1.6	For a CG amplifier: gm=5 mA/V, load RL=5kΩ, ro=80kΩ. Find voltage gain and input impedance.	02	2	2
1.7	Differentiate Class A and Class B stages with respect to their conduction angle and efficiency.	02	2	2

1.8	A class B complementary symmetry power amplifier using BJT s is operating from a single supply of 14V and driving a loud speaker load of 5Ω.The maximum power output possible is = _____.	01	1	1
1.9	A dual supply of 12V is used in a complementary symmetry class B power amplifier with load of 8Ω.The maximum power output possible=_____.	01	4	1
1.10	A power transistor with T_j(max)=200°C has θ_{JA}=5°C/W. P_d(max) for the transistor at an ambient temperature of 50°C= _____.	01	4	1
1.11	An op-amp with finite loop gain of 600 when connected as unity gain amplifier, it yields a gain of _____	01	1	1
1.12	Draw a super diode circuit suitable for rectification	02	3	2

Part B

Question No	Question	M	CO	BT
2a	In the circuit shown in Fig, draw the small signal model and determine V_0/V_S, Rin and Rout. (I_E=1.147mA). 	08	3	3
2b	Draw the small signal equivalent of the circuit shown in Fig and obtain the expression for the voltage gain, input impedance, output impedance. 	08	4	4
3a	For the amplifier shown in Figure below, find aspect ratio to achieve a voltage gain of 5 and output impedance of 1KΩ. The transistor is biased such that it operates 100mV away from the triode region. The capacitors used are very large and Gate resistance used is 1MΩ. Let kn'=200uA/v2 and λ=0.5 	08	3	3
3b	In the circuit of NMOS given below, VDD=16V, VG=2V,Vt=1 , K_n' W/L=2mA/V^2 find the value of R_D. Also find branch current through RL 	08	4	4

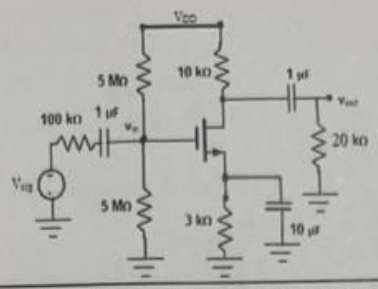
OR

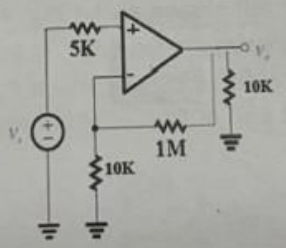
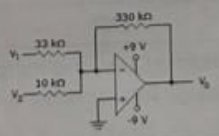
4a	Find the maximum value of V_G in the circuit of figure below, so that the MOSFET operates in saturation. $\mu_n C_{OX}$=200μA/V^2, W/L=2/0.2, V_t=0.5V, λ=0.	06	1	3
4b	The NMOS and PMOS transistors in the circuit given below are matched with kn'(W/L)n = kp'(W/L)p = 200uA/v2 and Vtn = -Vtp= 1V. Assuming λ=0 for both devices, find the drain currents i_{DN} and i_{DP} and the voltage vo for v_I = 0V, 2.5 V and -2.5 V. 	10	3	5
5a	Obtain expressions for voltage gain Vo/Vsig, Input and output impedance of a common gate configuration by considering channel length modulation.	10	3	4
5b	A student has rigged up the circuit shown in the Fig with C_{C1}=C_{C2}=0.01μF, C_S=10μF. Transistor V_t=1V, Kn'(W/L)=0.5mA/V^2, r_0=23.8KΩ and g_m=1.45mA/V. The Cgs and Cgd was measured as follows: C_{gs}=2pF and C_{gd}=0.4pF. Estimate the higher cut off frequencies due to input and output capacitance. 	06	2	3
	OR			
6a	Draw the frequency response of Common Source amplifier and derive expressions for lower cutoff frequency and upper cutoff frequency.	06	4	2
6b	In the CS amplifier circuit of Fig , given that C_{gs} = 1 pF, C_{gd} = 0.2 pF, $\mu_n C_{ox}(W/L)$ = 1mA/V^2 , λ = 0, I_D = 1mA,V_t = 1V, determine mid frequency gains v_{out} / v_{sig} & v_{out} / v_{in}	10	4	3

i) Lower cut off frequencies due to coupling and bypass capacitors

ii) Upper cutoff frequencies due to input and output capacitances

iii) 3 dB bandwidth



7a	Derive and compare the efficiencies of transformer coupled class A amplifier and Class B power amplifier.	06	4	2
7b	Prove that the maximum conversion efficiency in a complementary symmetry class B power amplifier is 78.5%.	06	1	1
7c	A power transistor for which the maximum junction temperature (T_{jmax})=150°C can dissipate maximum power Pd(max) of 40 W. Find θ_{JA} at ambient temperature TA=120°C, If TA=40°C, What will be the value of Pd?	04	4	4
	OR			
8a	An amplifier has a voltage gain of 500 without feedback. If a negative feedback is applied, the gain is reduced to 100. Calculate the fraction of the output fed back. If, due to ageing of components, the gain without feedback falls by 20%, calculate the percentage fall in gain with feedback.	06	2	4
8b	In the amplifier circuit of Fig, determine Vo/Vs, gain with feedback, Rin and Rout using feedback methodology. The opamp has an open loop gain 10^3, open loop input impedance and output impedance of 100kΩ and 1KΩ respectively. 	10	4	3
9a	Design a Schmitt trigger with UTP and LTP as 3V and 1V respectively. Let V_{CC}=12V and -V_{CC}=-12V.	06	3	3
9b	Find the output voltage Vo in terms of V1 and V2 for the circuit shown below. 	06	2	2
9c	Draw the circuit of Instrumentation amplifier and derive an equation for voltage gain.	04	3	2
	OR			
10a	Design a schmitt trigger to obtain UTP=-3V and LTP=2V. Assume VCC=-VEE=12V.	06	2	4
10b	Design a difference amplifier to obtain Rid=15KΩ, Acm=0, Ad=40dB. Find the worst case CMRR if resistors are 2% tolerance.	10	4	4